TIPA Korea Technology and Information Promotion Agency for SMEs

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**Developing Interpretable and Robust AI Systems for Regulation-Ready Life Science Applications**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (X), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (X), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Bio Medical			
	Medium Category	Small Category	Medium Category	Small Category
	Therapeutic Devices & Diagnostic Devices	Intelligent Interpretation Systems		
Concept of Proposed Technology	Machine learning and artificial intelligence are increasingly embedded in life science, medical devices, and digital health products. However, many AI systems currently fail to reach commercialization because their internal decision-making processes remain opaque, difficult to validate by domain experts, and insufficiently aligned with regulatory expectations. In regulated life science domains, it is not enough for an AI model to achieve high predictive performance; stakeholders such as biologists, clinicians, and regulators must be able to understand, trust, and validate how and why the model produces its outputs. The proposed technology addresses this gap by developing **interpretable, robust, and regulation-ready AI solutions tailored for life science applications**. Our approach focuses on lifting the "black box" nature of modern machine learning models while simultaneously increasing their robustness and stability (1). This enables domain experts to critically assess whether the model's behavior is biologically plausible, clinically meaningful, and safe for real-world use. Target use cases include clinical risk prediction, diagnostic imaging decision support, biomarker discovery, and patient stratification, with model behavior designed to be both expert-interpretable and robust under real-world data variability.			
Rationale for Joint R&D Project	The successful development of interpretable, robust, and regulation-ready AI systems in the life science domain requires close and continuous collaboration between AI technology developers and domain-specific industrial partners. Life science applications are characterized by complex data, strict regulatory requirements, and high safety expectations, which cannot be fully addressed through generic or off-the-shelf machine learning solutions.			

	A joint R&D project with Korean SMEs is relevant because the effectiveness and regulatory acceptance of the proposed AI technologies depend strongly on the **specific use case, data characteristics, clinical or biological context, and intended mode of deployment**. Korean SME partners contribute important domain expertise, real-world datasets, and a deep understanding of the practical and regulatory constraints within their target markets, such as medical devices, digital therapeutics, or biotechnology applications. This collaborative approach enables the systematic integration of **biological knowledge, expert feedback, and regulatory considerations** directly into the AI development process, rather than addressing these aspects retrospectively. Through iterative development and validation cycles, the joint R&D project reduces technical and regulatory risk while accelerating the transition from research prototypes to deployable AI components. Ideal Korean SME partners: (i) developing AI-enabled medical devices, digital therapeutics, or biotech analytics; (ii) possessing real-world biomedical data and a clear product pathway where interpretability and robustness are critical for regulatory approval.
Key Words	1. Interpretable AI / Explainable AI (XAI) 2. Robust Machine Learning 3. Regulation-Ready AI (MFDS / Global) 4. Clinical Decision Support 5. Biologically Informed Neural Networks

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level 4	End of the Project	Level 6
R&D Objectives	The primary objective of this joint R&D project is to develop and validate a suite of **interpretable, robust, and regulation-ready AI solutions** tailored to life science and healthcare applications. Building on existing research prototypes, the project aims to advance these technologies toward industrial-grade implementations that can be integrated into real-world products of Korean SMEs. Specifically, the project seeks to address two critical challenges commonly encountered in life science AI systems: 1. the lack of transparency and interpretability required for expert validation and regulatory approval, and 2. insufficient robustness of deep learning models, where small variations in input data may lead to unstable or unsafe predictions. Through joint development with Korean SME partners, the project aims to co-design AI models whose internal representations and decision mechanisms can be understood and evaluated by domain experts such as biologists and clinicians, while simultaneously ensuring stable and predictable behavior under realistic operating conditions. The overarching goal is to enable Korean SMEs to transform high-performing AI models into trustworthy, certifiable, and deployable components suitable for regulated life science markets.			
R&D Target Specifications	The target outcome of the project is a modular and extensible suite of AI methods and implementation components with the following characteristics: - **Interpretability**: AI models whose structure and outputs can be meaningfully interpreted and assessed by domain experts, supported by biologically informed architectures and structured learning approaches. - **Robustness and Stability**: Neural network models with controlled smoothness and stable input–output behavior, reducing sensitivity to noise, distribution shifts, or minor perturbations in the data. - **User-Controlled Trade-offs**: Mechanisms that allow users to explicitly balance predictive performance, interpretability, and robustness according to application-specific and regulatory requirements. - **Multi-modal Data Support**: Capability to handle complex life science data types, including clinical data, biological measurements, and imaging data, potentially combined within unified models. - **Industrial-Grade Implementation**: Well-documented, reproducible, and maintainable software components suitable for integration into SME development pipelines and products.			

R&D Plan	The R&D activities will be structured into iterative and collaborative phases, allowing flexibility to adapt to different SME use cases while maintaining a clear development trajectory: 1. **Use Case Definition and Requirement Analysis** Joint definition of the target application with the Korean SME partner, including data characteristics, intended usage scenarios, performance targets, and regulatory constraints. This phase establishes the technical and regulatory baseline for development. 2. **Data Processing and Preparation** Design and implementation of robust, reproducible, and well-documented data processing pipelines tailored to the selected use case. The goal of this phase is to ensure that the input data is processed in a transparent and consistent manner, providing a reliable foundation for machine learning model development and facilitating reproducibility, validation, and regulatory review. 3. **Model Design and Architecture Adaptation** Co-design of interpretable and robust AI model architectures tailored to the selected use case. This includes integrating biological prior knowledge, selecting appropriate robustness constraints, and defining interpretability mechanisms aligned with expert needs. 4. **Implementation and Prototype Development** Development of prototype AI models and software components, following best practices in software engineering, documentation, and reproducibility. Continuous feedback from the SME partner is incorporated throughout this phase. 5. **Validation and Robustness Evaluation** Systematic evaluation of model performance, interpretability, and robustness using real-world datasets provided by the SME partner. Expert feedback from biologists or clinicians is used to assess whether model behavior is meaningful and trustworthy. 6. **Integration and Demonstration** Integration of the developed AI components into the SME's existing workflows or prototype systems. Demonstration of the technology in a realistic usage scenario, supporting progression toward pilot-scale deployment. This phased and iterative R&D plan ensures close alignment between technological development, industrial relevance, and regulatory readiness, while enabling efficient knowledge transfer between partners.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The research institute will take primary responsibility for the design, development, and validation of the proposed AI technologies. In close collaboration with the Korean SME partner, the institute will: - Analyze the target use case and technical, operational, and regulatory requirements provided by the SME partner - Co-design interpretable and robust AI model architectures tailored to the specific life science application - Develop and implement high-quality, well-documented, and reproducible AI software components following industrial best practices - Lead the evaluation of model performance, interpretability, and robustness, incorporating feedback from domain experts - Provide technical leadership throughout the project, ensuring alignment with agreed objectives, milestones, and quality standards - Facilitate knowledge transfer to the Korean SME partner through documentation, technical discussions, and joint reviews
Expected R&R of Korean SME Partner	The Korean SME partner will play a key role in ensuring the industrial relevance and practical applicability of the developed AI solutions. The SME partner is expected to: - Define the concrete application scenario and business objectives for the joint R&D project - Provide access to relevant datasets and domain knowledge required for model development and validation - Specify functional, performance, and regulatory requirements, including constraints related to product deployment and certification - Actively participate in regular project meetings and reviews, providing feedback to guide model design and refinement - Support validation of the developed AI solutions in realistic usage scenarios and, where applicable, pilot-scale environments

5. Expected Outcomes/Impact

Expected Outcomes and Impact	The proposed joint R&D project is expected to generate significant technological, industrial, and strategic impact for Korean SMEs operating in life science, medical devices, and digital health sectors. First, the project will enable Korean SMEs to develop **interpretable and robust AI systems** that can be understood and validated by domain experts such as biologists and clinicians. This directly supports the establishment of trust in AI-based products, which is a critical prerequisite for adoption in clinical and life science environments. Second, the project will support **regulatory readiness and approval processes** by embedding interpretability, robustness, and validation mechanisms directly into the AI development lifecycle. By aligning AI system design with regulatory expectations, including those of Korean authorities such as the Ministry of Food and Drug Safety (MFDS) as well as international frameworks, the project will reduce regulatory risk and accelerate time-to-market for Korean SMEs. Third, participating SMEs will benefit from **technology transfer and capability building**, gaining access to advanced AI methodologies, high-quality and well-documented software implementations, and interdisciplinary expertise at the interface of artificial intelligence and life sciences. This will strengthen the SMEs' internal R&D capabilities beyond the duration of the project. Finally, at a broader level, the project will contribute to strengthening the **competitiveness of the Korean life science and healthcare industry**, enabling SMEs to develop globally competitive, trustworthy, and regulation-compliant AI-based products. By facilitating successful commercialization of AI technologies in regulated markets, the project supports Korea's strategic goals in advanced healthcare, digital innovation, and high-value technology exports.

6. References

(1) Hartman, Erik, et al. "Interpreting biologically informed neural networks for enhanced proteomic biomarker discovery and pathway analysis." Nature Communications 14.1 (2023): 5359.

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization